

Yuecheng XU

+86-180-6716-6890 | sayyx7@nottingham.edu.cn | xuyuecheng365@hotmail.com | yorking6.github.io

EDUCATION

University of Nottingham Ningbo China (QS #97) <i>Bachelor of Engineering in Architectural Environment Engineering</i>	Sept. 2024 – Jul. 2028 <i>Ningbo, China</i>
Tsinghua University - Technical University of Munich <i>Winter School Program in Smart Sustainable Cities and Energy</i>	Nov. 2025 – Dec. 2025 <i>Beijing, China</i>
Shanghai Jiao Tong University <i>Summer School Program in Net-Zero Carbon Fuels</i>	Jun. 2025 – Jul. 2025 <i>Shanghai, China</i>

EXPERIENCE

Energy Management System Developer <i>Green Carbon Intelligence (Ningbo) Technology Development Co., Ltd.</i>	Sept. 2025 – Present <i>Ningbo, China</i>
• System Development & Algorithm Optimization: Deeply involved in the R&D of an energy and carbon management system powered by deep learning and digital twin technologies; optimized energy dispatching algorithm logic to enhance the SaaS platform's intelligent responsiveness across diverse scenarios; the system has been successfully deployed by over 10 large-scale manufacturing clients, driving an average annual carbon reduction of 22% and an energy efficiency improvement of 18%. • Energy Auditing & ESG Consulting: Led or participated in 6 energy auditing and carbon emission accounting projects; conducted greenhouse gas (GHG) inventories based on the GHG Protocol and ISO 14064 standards; established energy consumption baselines and identified critical nodes for energy saving, providing core data support for clients' energy efficiency assessments and ESG disclosures. • Industry Insights & Deep Research Reports: Continuously tracked macro policies, cutting-edge technologies (such as carbon capture and green power trading), and market competitive dynamics in the energy and carbon management sector; conducted supply chain research and competitor analysis to independently or collaboratively compile 12 industry research reports, delivering strategic recommendations for product iteration and market expansion.	
New Energy Economics Consulting Specialist <i>CCID Consulting Co., Ltd. (State-Owned Enterprise under MIIT)</i>	Jun. 2025 – Sept. 2025 <i>Beijing, China</i>
• Market Sizing & Industrial Research: Focused on the new energy and intelligent equipment sectors, leading the authorship of 12 market sizing and industrial thematic studies; utilized top-down analysis to quantitatively evaluate market spaces, supply chain value distribution, and industry growth drivers for niche sectors, with core research findings integrated into the corporate industry database. • In-Depth Research Report Writing: Contributed to the formulation of 8 mid-year industrial research reports; conducted comprehensive reviews of macro policy shifts, technological iteration paths, and typical corporate case studies to deliver core conclusions on industry lifecycles and future trends, providing think-tank support for government decision-making and corporate strategic planning. • Database Construction & End-to-End Management: Managed the underlying maintenance of the energy and intelligent equipment database, collecting, cleaning, and validating over 20,000 data points; established data indexing standards through logical verification and dimensionality reduction, significantly accelerating the team's subsequent efficiency in data retrieval and chart generation.	
Consulting Assistant <i>Ningbo Zhenguo Bookstore Co., Ltd.</i>	Feb. 2025 – Jun. 2025 <i>Ningbo, China</i>
• Business Process Mapping & Lightweight Tool Development: Utilized Visio to map out operational business processes, successfully identifying efficiency bottlenecks and optimizing operational logic; spearheaded the development of two lightweight management tools—an automated Excel template library and a simplified knowledge graph—effectively boosting daily information retrieval and task processing efficiency by standardizing underlying data structures. • Knowledge System Construction & Product Design: Actively participated in the transition toward knowledge productization, taking charge of aggregating, integrating, and verifying massive amounts of industry information; categorized and analyzed user pain points and operational scenarios to build and maintain a standardized Frequently Asked Questions (FAQ) knowledge base, ensuring high accuracy and consistency in knowledge outputs.	

PROJECTS

AI-driven Optimization of Energy Systems |

Jul. 2025 – Present

- A high-impact research project focused on advancing the operational efficiency of district-scale thermal energy systems through sophisticated dynamic modeling and AI-driven control strategies.
- Dynamic System Modeling: Developed and calibrated a comprehensive, high-fidelity dynamic model of a complex district thermal energy system to accurately simulate transient behavior and predict future states.
- Advanced Control Implementation: Designed and deployed a data-driven optimization framework leveraging Model Predictive Control (MPC) and Machine Learning (ML) predictors for optimal energy management.
- Efficiency Enhancement: Successfully executed dynamic optimization strategies to minimize operational costs and maximize energy efficiency, demonstrating an improvement of 15% in system performance.
- Research Output: Authored a peer-reviewed quality manuscript detailing the novel methodology and empirical results.

Model-based Optimal Control of Industrial Energy Systems |

Mar. 2025 – Jun. 2025

- Conducted an extensive literature review on data-driven and physics-based modelling of integrated energy systems, with a focus on energy-efficient control frameworks and related methodologies.
- Assisted with industrial data mining, energy system modeling using Modelica, and other preparatory tasks.
- Primarily responsible for collecting and processing carbon emission factors, establishing time-series datasets, and applying Python for data analysis, dynamic modelling, and forecasting to support the development of control optimization strategies for emission reduction in industrial energy systems.
- Completed review paper on the development of dynamic carbon emission factors and generated data mining insights on relationship between renewable energy system capacity thresholds and environmental variables.

SCHOLARSHIP & AWARD

Provost Academic Scholarship (Second Level)

University of Nottingham Ningbo China

AY 2024 – 2025

Dec. 2025

TECHNICAL SKILLS

- **Programming & Tools:** Python, MATLAB, MySQL, C++, HTML/CSS, Git, VS Code, PyCharm
- **AI & Data Analysis:** Machine Learning, Reinforcement Learning, Large Language Models, Model Predictive Control, pandas, NumPy, Matplotlib, Data Visualization, Statistical Analysis
- **Energy Simulation & CAD:** Modelica, EnergyPlus, SIMIENS, HAP-Carrier, AutoCAD, Revit, Visio
- **Engineering & Standards:** Building Environment, Thermal System Modeling, GHG Protocol, ISO 14064, ESG Reporting, Carbon Accounting, Energy Auditing